

Full Stack Development with MERN

Nutrition Assistant Application

1. Introduction

Project Title

Nutrition Assistant Application

Team Members

Team Member	Role
Jahnvi Kanithi	Frontend Developer
Sai Bhargavi Yelagada	Backend Developer
Leela Mohan Banala	Database Developer
Narsipuram Avinash	Backend Integration & API Developer
Chandrakala Baswa	Testing, Documentation & Project Coordinator

2. Project Overview

Purpose

The **Nutrition Assistant Application** is a full-stack web application developed using the **MERN Stack (MongoDB, Express.js, React.js, and Node.js)** to help users maintain a healthy lifestyle through personalized nutrition management. The application enables users to track daily calorie intake, monitor nutritional progress, receive customized meal plans, and achieve their health and fitness goals.

The platform also provides dedicated dashboards for **Dietitians** and **Administrators**. Dietitians can manage client profiles, create personalized meal plans, and monitor progress, while administrators manage users, dietitians, and overall platform operations.

Objectives

- Promote healthy eating habits.
- Simplify nutrition and calorie tracking.
- Enable personalized meal planning.
- Improve communication between users and dietitians.
- Provide a secure and scalable nutrition management platform.

Key Features

User Module

- User Registration
- Secure Login
- Profile Management
- Daily Meal Tracking
- Nutrition Analysis
- Progress Monitoring
- Dashboard

Dietitian Module

- Client Management
- Meal Plan Creation
- Meal Plan Assignment
- Client Progress Monitoring

Administrator Module

- User Management
- Dietitian Approval
- Report Generation
- Platform Monitoring

Security

- JWT Authentication
- Password Encryption using bcrypt
- Role-Based Authorization

3. Architecture

Frontend Architecture

The frontend is developed using **React.js**, providing a responsive Single Page Application (SPA). React Router manages navigation, Axios handles communication with backend APIs, and Bootstrap is used for responsive UI design.

Technologies

- React.js
- HTML5
- CSS3

- JavaScript (ES6)
 - Bootstrap
 - Axios
 - React Router
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Backend Architecture

The backend is built using **Node.js** and **Express.js** following RESTful architecture.

Main backend modules include:

- Authentication Module
 - User Module
 - Client Management Module
 - Meal Plan Module
 - Nutrition Module
 - Progress Tracking Module
 - Admin Module
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Database Architecture

MongoDB stores application data using Mongoose ODM.

Collections:

- Users
- Clients
- MealPlans
- Progress

Relationships

- One Dietitian → Multiple Clients
 - One Client → Multiple Meal Plans
 - One User → Multiple Progress Records
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4. Setup Instructions

Prerequisites

- Node.js (v16 or above)
- npm
- MongoDB Community Server or MongoDB Atlas
- Visual Studio Code
- Git
- Postman

Installation

Clone Repository

```
git clone <repository-url>
```

Frontend

```
cd client
```

```
npm install
```

Backend

```
cd server
```

```
npm install
```

Environment Variables (.env)

```
PORT=5000
```

```
MONGO_URI=mongodb+srv://pallerladhanunjayareddy54_db_user:DhanuMongo123@cluster0.j84v  
tai.mongodb.net/nutrition_app?retryWrites=true&w=majority
```

```
JWT_SECRET=supersecretkey123
```

5. Folder Structure

Nutrition-Assistant/

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| └─ client/

| └─ public/

| └─ src/

| | └─ components/

| | └─ pages/

| | └─ services/

| | └─ context/

| | └─ App.js

| | └─ index.js

|

└─ server/

| └─ config/

| └─ controllers/

| └─ middleware/

| └─ models/

| └─ routes/

| └─ utils/

| └─ server.js

|

└─ package.json

└─ README.md

6. Running the Application

Frontend

cd frontend

npm start

Backend

cd backend

npm start

7. API Documentation

Authentication

Method	Endpoint	Description
POST	/api/auth/register	Register User
POST	/api/auth/login	User Login

Users

Method	Endpoint	Description
GET	/api/users/profile	View Profile
PUT	/api/users/profile	Update Profile

Clients

Method	Endpoint	Description
GET	/api/clients	View Clients

POST	/api/clients	Add Client
PUT	/api/clients/:id	Update Client
DELETE	/api/clients/:id	Delete Client

Meal Plans

Method	Endpoint	Description
GET	/api/mealplans	View Meal Plans
POST	/api/mealplans	Create Meal Plan
PUT	/api/mealplans/:id	Update Meal Plan
DELETE	/api/mealplans/:id	Delete Meal Plan

Progress

Method	Endpoint	Description
GET	/api/progress	View Progress
POST	/api/progress	Add Progress

8. Authentication

The application implements secure authentication using **JSON Web Tokens (JWT)**.

Authentication Process

1. User registers.
2. Password is encrypted using **bcrypt**.
3. User logs in.
4. JWT token is generated.
5. Token is stored on the client.
6. Protected APIs verify the token before allowing access.

Authorization

Three user roles are supported:

- User
- Dietitian
- Administrator

Each role has access only to authorized resources.

9. User Interface

The application includes the following pages:

- Home
- Register
- Login
- User Dashboard
- Dietitian Dashboard
- Admin Dashboard
- Profile
- Client Management
- Meal Planning
- Nutrition Analysis
- Progress Tracking
- Reports

10. Testing

Testing was performed using:

- Manual Functional Testing
- API Testing using Postman
- CRUD Testing
- Authentication Testing
- Input Validation Testing
- Cross-Browser Testing
- Responsive UI Testing

11. Screenshots / Demo

Include:

- Home Page
- Login Page
- Registration Page

- Dashboard
 - Meal Plan Module
 - Client Management
 - Progress Charts
 - Nutrient Analysis
 - Admin Dashboard
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12. Known Issues

- AI-based meal recommendations are not included in the current version.
 - Push notifications are not implemented.
 - Offline functionality is not supported.
 - Third-party wearable device integration is not available.
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13. Future Enhancements

- AI-powered personalized meal recommendations.
 - Mobile application using React Native.
 - Integration with fitness wearables (Fitbit, Google Fit).
 - Barcode scanning for food items.
 - Voice-based meal logging.
 - Video consultation with dietitians.
 - Multi-language support.
 - Email and push notifications.
 - Cloud deployment with auto-scaling.
 - Advanced analytics and predictive nutrition insights.
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GitHub link : <https://github.com/leela-mohan123/Nutrition-Assistant>

Demo video links :

<https://drive.google.com/drive/folders/10NmImDog2tzFmfMtLpvc70OKS8fiQYIG>